

Gabriel P. Oliveira

Data Scientist and PhD Candidate

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Education

- 2021–Present **Ph.D. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Visiting student at **Politecnico di Torino, Italy** (2024–25)
Thesis: On the dynamics of music virality and its relation with mainstream success
Advisor: Prof. Mirella M. Moro, Ph.D. **Current GPA:** 94.9 out of 100
Co-advisor: Prof. Ana Paula Couto da Silva, Ph.D. **Expected Graduation:** October 2025
- 2019–2021 **M.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Thesis: Analyses of musical success based on time, genre and collaboration **GPA:** 92.0 out of 100
Advisor: Prof. Mirella M. Moro, Ph.D.
Co-advisor: Prof. Anisio Lacerda, Ph.D.
- 2015–2018 **B.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Tie Strength in GitHub Heterogeneous Networks **GPA:** 91.3 out of 100
Advisor: Prof. Mirella M. Moro, Ph.D.
Co-advisor: Prof. Michele A. Brandão, Ph.D.
- 2012–2014 **High School Technical Degree in Computer Networks**, *Centro Federal de Educação Tecnológica de Minas Gerais (CEFET-MG)*, Brazil.

Professional Experience

- 2021–2024 **Data Scientist**, *Analytical Capabilities Project*, UFMG/MPMG.
Member of the Analytical Capabilities Project, a collaboration between the Computer Science Department of UFMG and the Prosecution Service of the State of Minas Gerais (MPMG). My activities included building models for identifying fraud in public bidding processes and using machine learning and natural language processing techniques to classify and enhance public expenditure data.
- 2019–2023 **Research Assistant**, *Project Både*, UFMG.
The goal of this project is to develop semantic metrics derived from social network data to enhance several applications and services. Specifically, my research used social network analysis and data mining techniques to analyze different aspects of musical success and how they relate to collaboration and music genres.
- 2019 **Data Science Intern**, UFMG/Localiza.
As a member of the Data Science Residency Program, a collaboration between the Computer Science Department of UFMG and Localiza, I engaged in developing machine learning models to address practical challenges within the car rental industry. My responsibilities involved optimizing marketing operations within this sector by using techniques such as clustering and association rule mining.
- 2018–2019 **Software Engineering Intern**, dti digital crafters.
In this role, I focused on modeling SQL databases, crafting APIs using C# .NET, and designing interfaces with HTML5/CSS3 and AngularJS.
- 2017–2018 **Undergraduate Research Assistant**, *Project FoRGit*, Universidade Federal de Minas Gerais.
The goals of this project included building a heterogeneous social collaboration network from GitHub, analyzing new parameters that measure the interactions between GitHub developers, and propose new metrics for their social capital.
- 2016–2017 **Undergraduate Teaching Assistant**, *Computer Science Department*, UFMG.
Teaching assistant in the Computer Programming course (DCC001) for Engineering students.
- 2015 **IT Support Intern**, *Support Center for e-learning*, UFMG.
My activities included network and server management, user support, support to audiovisual content production, and web conferencing management.

Skills

Technologies	Python, SQL, Jupyter, PySpark, C/C++, C#, JavaScript, Apache Hive
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, NetworkX
Hard Skills	Network Science, Machine Learning, Data Mining, Statistical Analysis
Soft Skills	Communication, Teamwork, Problem Solving, Career-focused

Languages

Portuguese	Native	
English	Advanced	TOEFL ITP score: 613 out of 677 (CEFR B2), 2018
Italian	Intermediate	
Spanish	Basic	

Awards

- 2023 **Best Master Thesis Award**, Brazilian Symposium on Databases (SBBD).
Prize awarded in the 5th Contest of Theses and Dissertations on Databases.
- 2022 **Best Master Thesis Award – Finalist**, Congress of the Brazilian Computer Society (CSBC).
One of the ten Master theses selected for presentation at the Theses and Dissertations Contest of the Congress of the Brazilian Computer Society.
- 2018 **Best Graduating Student Award – 1st Runner-up**, Universidade Federal de Minas Gerais.
Prize awarded for students with the highest GPA in the Computer Science Undergraduate Program.

Selected Publications

The complete list of publications is available on my webpage.

- 2024 **A quantitative comparison of viral and hit songs in the Brazilian music market**, *Vórtex Musical Journal*.
GP Oliveira, APC da Silva, MM Moro
- 2024 **Analyzing the temporal relation between virality and success in the Brazilian music market**, *Integrated Software and Hardware Seminar (SEMISH)*, Brasília, Brazil, **Best Paper Award**.
GP Oliveira, APC da Silva, MM Moro
- 2024 **What makes a viral song? Unraveling music virality factors**, *ACM Web Science Conference (WebSci)*, Stuttgart, Germany.
GP Oliveira, APC da Silva, MM Moro
- 2023 **Hot streaks in the music industry: identifying and characterizing above-average success periods in artists' careers**, *Scientometrics*.
GP Oliveira, MO Silva, DB Seufitelli, GRG Barbosa, BC Melo, MM Moro
- 2023 **Hit song science: a comprehensive survey and research directions**, *J. of New Music Research*.
DB Seufitelli, GP Oliveira, MO Silva, C Scofield, MM Moro
- 2020 **Detecting Collaboration Profiles in Success-based Music Genre Networks**, *International Society for Music Information Retrieval Conference (ISMIR)*, Montréal, Canada, **Best Paper Presentation Award**.
GP Oliveira, MO Silva, DB Seufitelli, A Lacerda, MM Moro

Other activities

- 2024 External Reviewer – SEMISH, DSW
- 2023 External Reviewer – BraSNAM, SEMISH, DSW
- 2022 External Reviewer – BraSNAM, SBBD
- 2021 External Reviewer – WebMedia, SEMISH, WEI